NUMERICAL RANGE OF A WEIGHTED COMPOSITION OPERATOR

ØISTEIN SØVIK

AI assistance was used in adapting and checking the argument and the exposition.

The source asks: "Does 0 lie in the interior of the numerical range $W(C_{\psi,\varphi})$?" Question 2.13, p. 8 of Lahiri, Ojha, Halder, and Birbonshi's paper on numerical ranges and Davis–Wielandt shells of composition operators on the Hardy space of Dirichlet series. The hypotheses are that φ is a nonconstant c_0-symbol, $\psi \in \mathscr{H}^2$, $\psi_1 = 0$, and $C_{\psi,\varphi}$ is bounded.

Lahiri, Ojha, Halder, and Birbonshi, *Numerical range and invariant subspaces of weighted composition operators on the Hardy space of Dirichlet series*, arXiv:2608.29294.

Here $C_{\psi,\varphi} f = \psi(f \circ \varphi)$, $\psi_1 = \psi(+\infty)$, and $W(T) = \{\langle Tx, x\rangle : \|x\| = 1\}$.

Answered: no as stated; yes for nonzero ψ. Evidence: Sketch.

No as stated: $\psi = 0$ gives numerical range $\{0\}$. If $\psi \neq 0$, the answer is yes.

Proof. Let $T = C_{\psi,\varphi}$ and $u = \psi/\|\psi\|_2$. Since $\psi_1 = 0$, 1 and u are orthonormal. Also $T1 = \psi$ and $\langle Tu, 1\rangle = 0$, because φ has positive characteristic and u has zero constant coefficient. Thus the compression of T to $\operatorname{span}\{1, u\}$ has matrix

$$\begin{pmatrix} 0 & 0 \\ \|\psi\|_2 & \lambda \end{pmatrix}, \qquad \lambda = \langle Tu, u\rangle.$$

Its numerical range is an ellipse whose foci are 0 and λ. Its minor and major semiaxes are respectively

$$\frac{\|\psi\|_2}{2} \quad \text{and} \quad \frac{\sqrt{\|\psi\|_2^2 + |\lambda|^2}}{2}.$$

Since 0 is a focus, the ellipse contains the disc centered at 0 of radius

$$\frac{\sqrt{\|\psi\|_2^2 + |\lambda|^2} - |\lambda|}{2} > 0.$$

It therefore contains a disc of positive radius centered at 0. The numerical range of a compression is contained in $W(T)$, proving the claim for nonzero ψ; for $\psi = 0$, $T = 0$ and $W(T) = \{0\}$. □

Date: 13 October 2026.